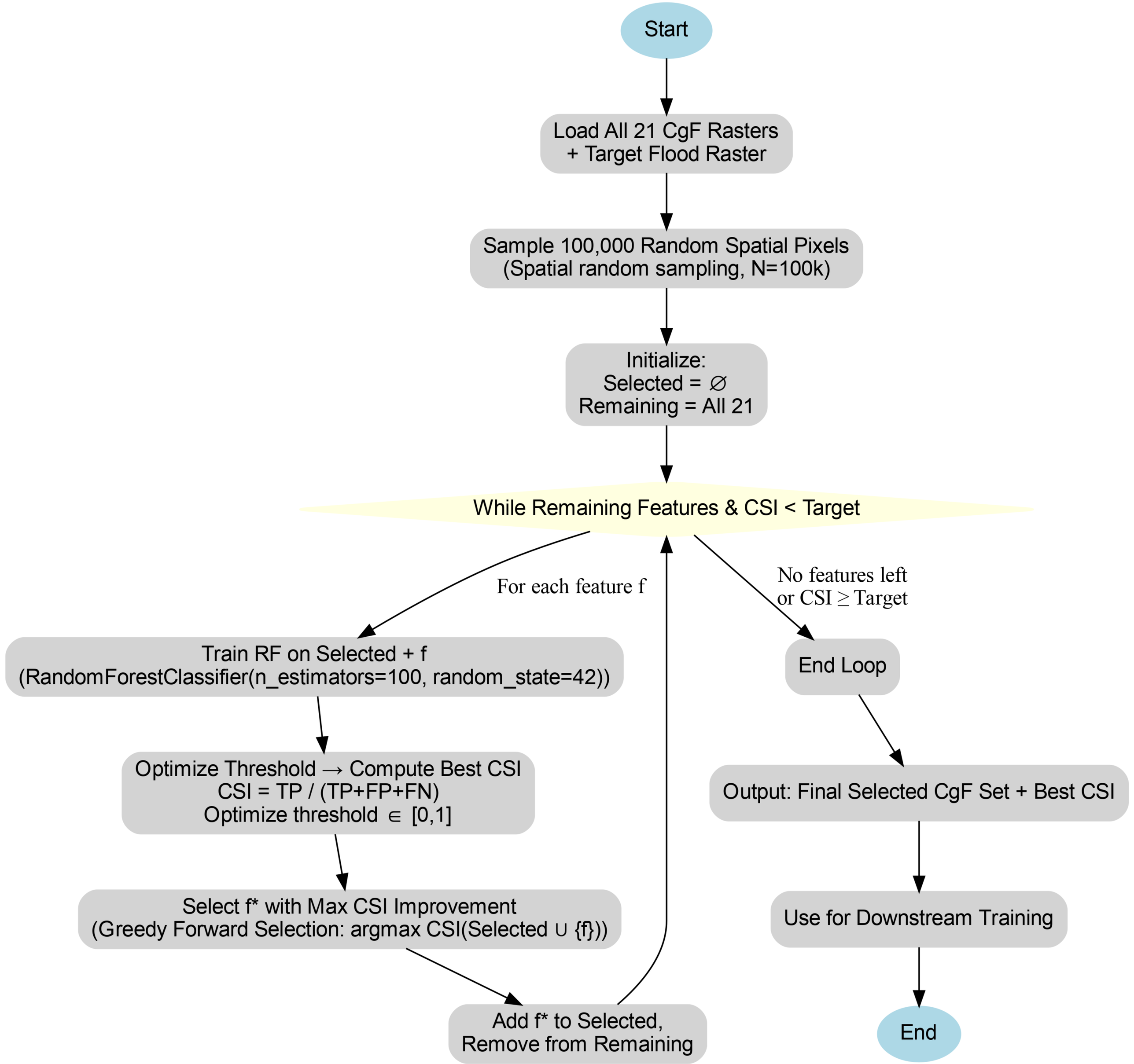
Start
Load All 21 CgF Rasters
+ Target Flood Raster
Sample 100,000 Random Spatial Pixels
(Spatial random sampling, N=100k)
Initialize:
Selected = ∅
Remaining = All 21
While Remaining Features & CSI < Target
For each feature f
No features left
or CSI ≥ Target
Train RF on Selected + f
(RandomForestClassifier(n_estimators=100, random_state=42))
Optimize Threshold → Compute Best CSI
CSI = TP / (TP+FP+FN)
Optimize threshold ∈ [0,1]
Select f* with Max CSI Improvement
(Greedy Forward Selection: argmax CSI(Selected ∪ {f}))
Add f* to Selected,
Remove from Remaining
End Loop
Output: Final Selected CgF Set + Best CSI
Use for Downstream Training
End